

Slide 0 - Title

Today we'll walk you through our project "Computer Vision - Multi-Label Image Classification: Query2Label and MLSPL". I'm Christine, presenting with Andreas and Riajul. We will go into why multi-label learning matters, outline two complementary methods - Query2Label for feature recognition from the paper "Query2Label: A Simple Transformer Way to Multi-Label Classification", and ROLE loss for sparse annotations from the paper "Multi-Label Learning from Single Positive Labels", and describe how we reproduced them on MS-COCO 2014 benchmark dataset, and summarise the results.

Multi-label Learning From Single Positive Labels

In multi-label classification, an image can belong to several categories. However, in the MLSPL setting, we only know one positive label per image. All other labels are unobserved—not confirmed negatives. This makes standard losses like Binary Cross-Entropy problematic, because they treat missing labels as negative, which introduces many false negatives and hurts performance.

Regularised Online Label Estimation (ROLE)

The authors of MLSPL propose a new approach called ROLE — Regularised Online Label Estimation — to directly address this. Instead of assuming unobserved labels are negative, they train two models jointly:

- A classifier f that makes predictions, and

- A label estimator g that assigns soft probabilities to the unknown labels.

They also include a regularization term that encourages each image to have a realistic number of positive labels. The training alternates between updating f and g , refining both over time.

The ROLE Loss Function

This idea is encoded directly in the loss function. The ROLE loss has two main parts: The LBCE term encourages the classifier predictions to match the estimated soft labels. The LEPR term ensures the model still predicts known positives, and it penalizes the model if it predicts too many or too few labels overall.

This formulation lets the model learn to recover missing labels without falsely treating them as negatives.

Final Loss Equation

These components come together in the final ROLE loss: It averages the classifier-to-estimator and estimator-to-classifier objectives. This symmetric design helps both models improve in tandem — refining predictions and label estimates together.

Experimental setup

For our experiments, we used two NVIDIA RTX 3060 GPUs and ran both Q2L and MLSPL implementations. We adapted the environment for Q2L due to some package mismatches, particularly for RandAugment. For MLSPL, we used a more recent PyTorch version and ran all experiments on a single GPU for consistency.

Dataset and metric

The dataset used was MS-COCO 2014, which includes over 120,000 images and 80 object categories. Each image can have multiple labels, making it ideal for multi-label classification. We report results using mean Average Precision (mAP), which is standard in this task.

Q2L backbones

We evaluated six backbones inside the Q2L framework: ResNet-101 at resolutions 448 and 576, TResNet-L with and without ImageNet-22k pre-training, and two transformer-style models: Swin-Large and CvT-w24. Due to time constraints, the evaluation is based on pretrained models made publicly available by the authors. However, the training of the backbones were initialized, which yielded memory overflow by the Swin and CvT architectures, even with the lowest batchsize. The other backbones could be trained with a lower batchsize.

MLSPL protocol

For MLSPL, we followed the paper’s protocol: We created a single positive label per image and trained a linear classifier for 25 epochs. Then we fine-tuned the full model for 10 more epochs. We also performed a hyperparameter search over learning rates and batch sizes, settling on optimal values for both phases.

Q2L results

Our Q2L results are shown here. For all tested backbones, our reproduced mAP scores match the original paper’s results closely — confirming the correctness of our setup.

MLSPL results

For MLSPL with the ROLE loss, our results also align well with the paper. We achieved 66.3% mAP using the linear model and slightly better — 66.9% — with end-to-end fine-tuning. These results show that ROLE is highly effective, especially given that it’s trained from just one positive label per image.